

Stationarity Balance Beats Population Balance

A Defect-Field Benchmark on Open Quantum Pointer States

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Abstract

This paper tests whether open quantum pointer robustness is better described by population balance or by stationarity balance. The benchmark is a classical Lindblad simulation, not a quantum-hardware experiment, a collapse model, or a derivation of quantum measurement.

We simulate 3600 one-qubit Lindblad trajectories across six channel families: z -dephasing, x -dephasing, amplitude damping, thermal relaxation, mixed z/x dephasing, and depolarizing noise. The target is a pointer-robustness score combining initial-state fidelity retention, measurement-probability stability, and final purity. The central comparison is between

$$B_{\text{pop}} = 1 - |p_0 - p_1|,$$

and stationarity balance,

$$B_{\text{stat}} = \frac{1}{1 + 2|\dot{p}|_0 + 0.35 \left| \frac{d}{dt} \text{Tr}(\rho^2) \right|_0}.$$

The result is unambiguous in this benchmark. Scalar stationarity substantially outperforms scalar population balance:

$$R_{\text{stat}}^2 = 0.6457, \quad R_{\text{pop}}^2 = 0.4561, \quad \Delta R^2 = +0.1896.$$

The full stationarity-sensitive defect-field model reaches $R^2 = 0.8527$ in ordinary cross-validation. A strong standard quantum baseline remains higher in ordinary cross-validation ($R^2 = 0.9011$), but the defect-field model performs best under leave-channel-family-out transfer ($R^2 = 0.7075$ versus 0.6637).

Thus the useful conclusion is narrow: in open quantum systems, balance coordinates for pointer robustness should be defined through generator stationarity, not through raw population equality.

1 Purpose

The purpose of this paper is to stress-test a simple point. In a quantum pointer-state setting, raw population balance is not the same as robustness. A pointer state such as $|0\rangle$ can be maximally stable while being maximally imbalanced in population. Therefore a balance coordinate of the form

$$B_{\text{pop}} = 1 - |p_0 - p_1| \tag{1}$$

is not automatically appropriate for pointer robustness.

The alternative tested here is stationarity balance:

$$B_{\text{stat}} = \frac{1}{1 + 2|\dot{p}|_0 + 0.35 \left| \frac{d}{dt} \text{Tr}(\rho^2) \right|_0}. \tag{2}$$

This quantity is high when the initial measurement probabilities and purity are locally stable under the open-system generator.

The paper asks:

Does stationarity-sensitive balance predict open-system pointer robustness better than raw population balance?

2 Lindblad Benchmark

The dynamics are generated by a standard one-qubit Lindblad equation:

$$\dot{\rho} = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right). \quad (3)$$

The Hamiltonian is

$$H = h_x \sigma_x + h_y \sigma_y + h_z \sigma_z. \quad (4)$$

The channel families include:

- z -dephasing;
- x -dephasing;
- amplitude damping;
- thermal relaxation;
- mixed z/x dephasing;
- depolarizing noise.

For each sampled initial pure state and generator, the trajectory is integrated classically by a fourth-order Runge–Kutta method. This is an entirely classical density-matrix simulation.

3 Pointer Robustness Target

For each generator, an effective environmental pointer axis $\hat{n}_{\text{ptr}}$ is constructed from the dominant noise channels. Let

$$p(t) = \frac{1}{2} (1 + \mathbf{r}(t) \cdot \hat{n}_{\text{ptr}}) \quad (5)$$

be the measurement probability along this axis.

The primary target is

$$\boxed{Y_{\text{ptr}} = 0.50 F_{\text{ret}} + 0.35 P_{\text{stab}} + 0.15 \text{Tr}(\rho_T^2)}. \quad (6)$$

Here

$$F_{\text{ret}} = \langle \psi_0 | \rho_T | \psi_0 \rangle, \quad (7)$$

$$P_{\text{stab}} = 1 - |p(T) - p(0)|. \quad (8)$$

This target is not a proposed universal definition of pointer states. It is a transparent benchmark target combining state survival, measurement-probability stability, and final purity.

4 Structural Features

The scalar population version uses

$$(H_{\text{stat}}, B_{\text{pop}}, S_{\text{coh}}, V_{\text{align}}, R_{\text{resp}}, R_{\text{rob}}, F_{\text{scalar}}) . \quad (9)$$

Here S_{coh} is coherent activity transverse to the pointer axis, and V_{align} measures alignment with that axis.

The stationarity version replaces B_{pop} by B_{stat} . The full defect-field model adds local generator pressures:

- initial Lindblad residual norm;
- initial pointer-probability drift $|\dot{p}|_0$;
- purity-decay pressure;
- channel conflict;
- damping and excitation pressure;
- decoherence exposure;
- transverse Hamiltonian pressure;
- pointer-alignment loss.

These features are all computed from the initial state and the generator, not from the final target.

5 Baselines and Protocol

The compared feature sets are:

- rates only;
- state geometry only;
- scalar population balance;
- scalar stationarity balance;
- full stationarity-sensitive defect field;
- a strong standard quantum baseline;
- shuffled defect-field null.

Models are evaluated with five-fold cross-validation using ridge regression and random forests. The stricter stress test is leave-channel-family-out transfer, where one Lindblad channel family is held out entirely.

6 Results

6.1 Cross-Validation

Table 1: Random-forest cross-validation results.

Feature set	R^2	RMSE	Spearman
Standard quantum baseline	0.9011	0.0362	0.9241
Defect-field stationary	0.8527	0.0442	0.8856
Scalar stationarity	0.6457	0.0686	0.7719
Scalar population balance	0.4561	0.0850	0.6221
Rates only	0.1641	0.1054	0.4646
State geometry only	0.0008	0.1152	0.2945
Shuffled defect null	-0.0261	0.1168	-0.0003

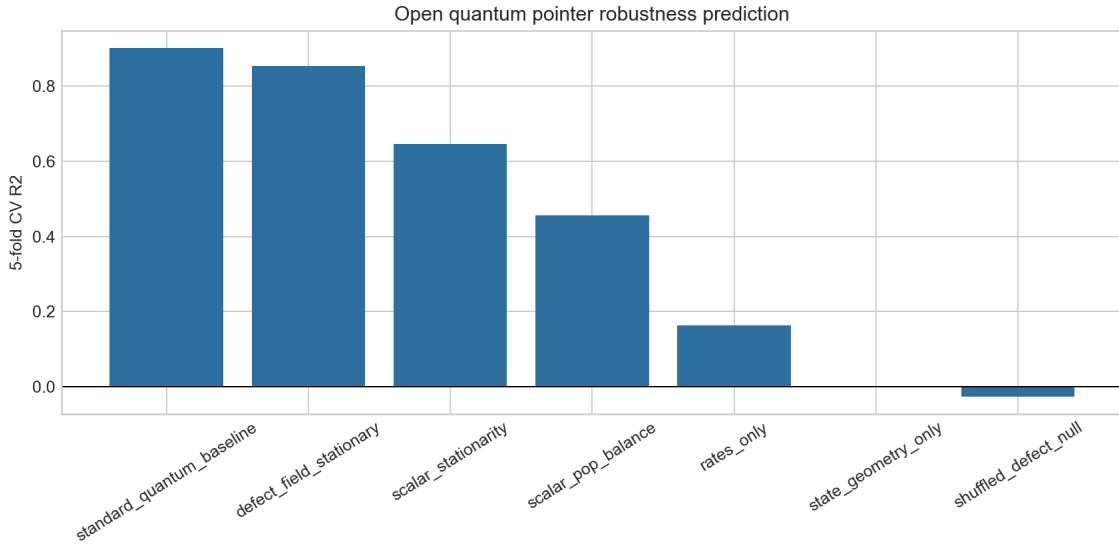


Figure 1: Cross-validation comparison. Stationarity-sensitive scalar and defect-field features substantially outperform raw population balance and shuffled nulls, but do not surpass the strongest standard quantum baseline in ordinary cross-validation.

6.2 Population Balance vs Stationarity Balance

The key ablation is:

$$\Delta R^2 = R^2_{\text{scalar stat}} - R^2_{\text{scalar pop}} = 0.6457 - 0.4561 = 0.1896. \quad (10)$$

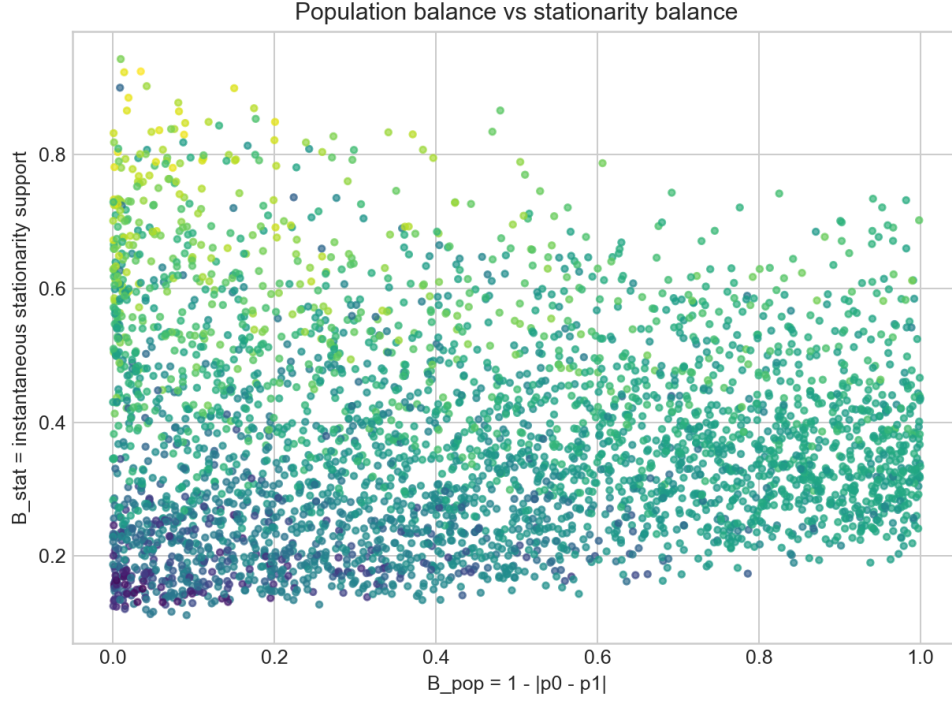


Figure 2: Raw population balance versus stationarity balance. Pointer-robust states need not be population-balanced; they must be stable under the generator.

6.3 Feature Importance

Table 2: Top stationarity-sensitive defect-field feature importances.

Feature	Permutation importance
$ \dot{p} _0$	0.7927
Damping pressure	0.3198
Channel conflict	0.1116
Pointer-alignment loss	0.0638
V_{align}	0.0540
Coherence exposure	0.0456
Purity-decay pressure	0.0438
Decoherence pressure	0.0424
B_{stat}	0.0298

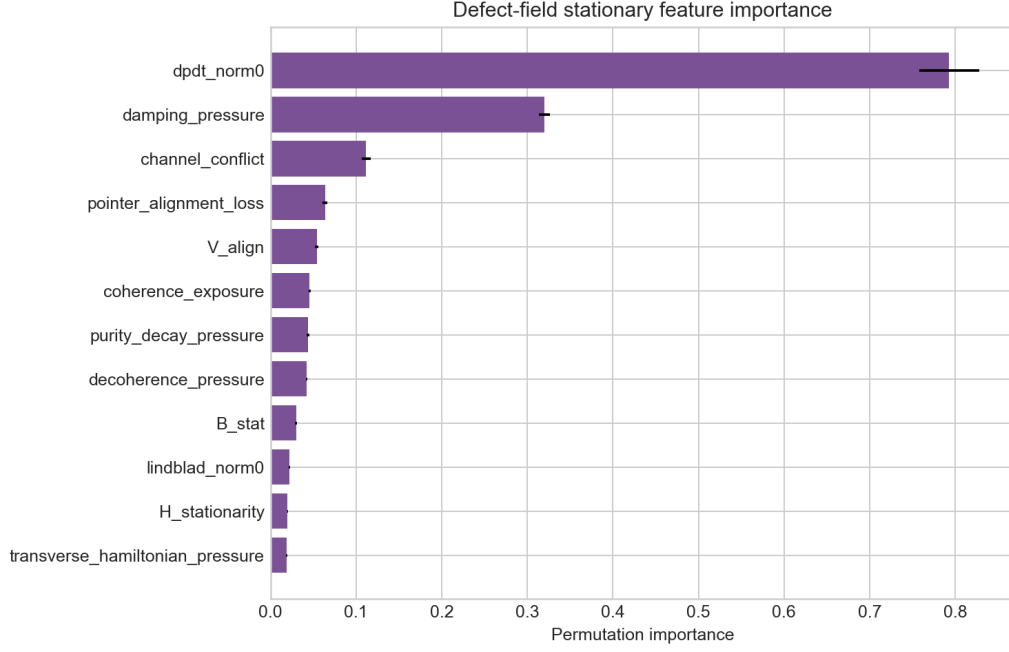


Figure 3: Permutation importance inside the stationarity-sensitive defect-field model. The strongest predictor is initial pointer-probability drift.

6.4 Leave-Channel-Family-Out Transfer

Table 3: Overall leave-channel-family-out results.

Feature set	R^2	RMSE	Spearman
Defect-field stationary	0.7075	0.0623	0.8286
Standard quantum baseline	0.6637	0.0668	0.8105
Scalar stationarity	0.4928	0.0821	0.7284
Scalar population balance	0.3108	0.0957	0.5604
Rates only	0.0006	0.1152	0.2324
State geometry only	-0.2561	0.1292	0.0087

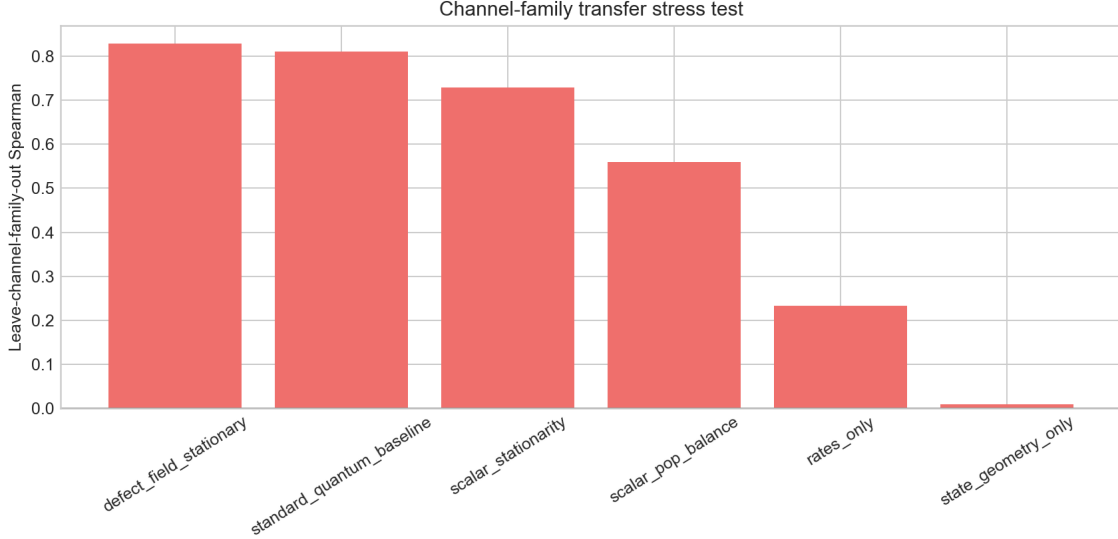


Figure 4: Leave-channel-family-out transfer. The stationarity-sensitive defect field is the best overall transfer model in this benchmark.

7 Interpretation

The result is not that structural selection beats quantum mechanics. The strongest ordinary cross-validation model is the standard quantum baseline, as it should be in a well-posed physical simulation.

The useful result is more precise:

For pointer robustness, balance should be defined through stationarity, not through raw population equality.

This resolves a concrete ambiguity. Raw population balance rewards equal populations, but pointer states are often unequal-population eigenstates of the environment. The stationarity definition instead asks whether the measurement probabilities and purity are locally stable under the generator. Methodologically, this matters because it shows that structural coordinates in open quantum systems cannot be imported from generic support language without checking what the dynamics actually preserve.

The leave-channel-family-out result is also important. Although the standard quantum baseline is stronger in ordinary cross-validation, the stationarity-sensitive defect field transfers better across held-out channel families in this benchmark. This suggests that stationarity pressure, damping pressure, channel conflict, and pointer alignment form a more portable structural description than raw population balance alone.

8 Limitations

- The experiment is a one-qubit toy simulation, not a many-body decoherence model.
- The pointer axis is an effective diagnostic construction from channel weights, not a microscopic theorem.
- The target combines fidelity, probability stability, and purity by declared weights; other pointer targets should be tested.
- No quantum hardware is used or required.

- The result does not derive the Born rule, solve the measurement problem, or constitute an experimental claim.

9 Reproducibility

Repository:

<https://github.com/Chris4081/structural-selection-principle>

The experiment is stored in:

`experiments/open_quantum_pointer_paper58/`

Run:

```
cd experiments/open_quantum_pointer_paper58
python3 open_quantum_pointer_paper58.py
```

Outputs are written to:

`outputs_paper58/`

No external quantum dataset is redistributed. All CSV/JSON/PNG files are generated locally by the script.

10 Conclusion

Paper 58 shows that open quantum pointer robustness can be studied without access to a quantum computer. A classical Lindblad benchmark is sufficient to test whether structural defect-field definitions are aligned with known decoherence behaviour.

The main conclusion is narrow but useful: stationarity-sensitive balance is substantially better than raw population balance for pointer robustness. This supports a practical refinement of the structural support language in quantum settings:

B_{quantum} should track stationarity, not mere population equality.

References

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